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SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1715

COURSE OUTCOME COVERED

***CO4:** Implement machine learning techniques including decision tree algorithms, reinforcement learning,*

and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 10%

Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

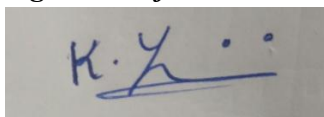
ARTICLE REVIEW

Code of Conduct:

I _K.Yamini(192572181)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'K. Yamini' with a stylized flourish at the end.

Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
- (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Task 1 (10)	Task 2 (10)	Task 3 (10)	Task 4 (10)	Total Marks (40)
Understanding of Article	20% (2)	/2	/2	/2	/2	/8
Summary	20% (2)	/2	/2	/2	/2	/8
Critical Analysis	25% (2.5)	/2.5	/2.5	/2.5	/2.5	/10
Use of Evidence	15% (1.5)	/1.5	/1.5	/1.5	/1.5	/6
Organization	20% (2)	/2	/2	/2	/2	/8
Total per Question	10 Marks	/10	/10	/10	/10	/40

ARTICLE REVIEW

Selected Research Article

Title: Fault Identification of Photovoltaic Array Based on Machine Learning Classifiers

Authors: Mohamed M. Badr, Mostafa S. Hamad, Ayman S. Abdel-Khalik, Ragi A. Hamdy, Shehab Ahmed, and Eman Hamdan

Journal: IEEE Access, Volume 9, pp. 159113–159132

Year: 2021

DOI: 10.1109/ACCESS.2021.3130889

Domain/Application Area: Solar photovoltaic (PV) systems, renewable energy, electrical fault detection and diagnosis

Machine Learning Problem: Classification of healthy and faulty photovoltaic-array operating conditions using supervised machine-learning classifiers.

TASK 1: ARTICLE SUMMARY

1(a) Full Citation, Domain and ML Problem

Full Citation

Badr, M. M., Hamad, M. S., Abdel-Khalik, A. S., Hamdy, R. A., Ahmed, S., & Hamdan, E. (2021). **Fault Identification of Photovoltaic Array Based on Machine Learning Classifiers**. *IEEE Access*, 9, 159113–159132. DOI: 10.1109/ACCESS.2021.3130889.

The article investigates the application of machine learning for **fault detection and diagnosis in photovoltaic arrays**. The study focuses on a 4 kW grid-connected photovoltaic system and considers several operating conditions, including healthy operation, arc fault, maximum power point tracking (MPPT) unit failure, line-to-line fault, open-circuit fault, and shading conditions.

The main machine-learning problem is a **multi-class classification problem**. The input data consist of electrical and environmental measurements obtained from the PV system. These measurements are used to determine whether the PV array is operating normally or experiencing a particular type of fault. The authors compare **Decision Tree (DT), K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)** classifiers. Different splitting criteria, distance functions, kernels, and hyperparameters are examined. Bayesian optimization is also used to identify suitable hyperparameters for the classifiers.

The article therefore directly relates to the CO4 topics of **Decision Trees, Statistical Learning Methods, and supervised/inductive learning**.

1(b) Key Objective, Research Gap and Proposed Solution

The main objective of the article is to develop an effective machine-learning-based fault detection and diagnosis framework for grid-connected photovoltaic arrays. Conventional protection devices such as overcurrent protection devices can detect some serious faults, but they may not provide sufficient information to distinguish between different fault types. Their performance can also be affected by solar irradiance, fault impedance, MPPT operation, and the location of the fault. Therefore, a more intelligent method is required to identify different PV fault conditions accurately and quickly.

The research gap identified by the authors is that existing protection approaches have limited diagnostic capability, while machine-learning studies may not systematically compare different classifier configurations under a wide range of PV operating and fault conditions. To address this gap, the authors developed a fault detection and diagnosis framework using Decision Tree, KNN, and SVM classifiers. They generated 14,100 samples from a MATLAB/Simulink PV-system model under different irradiance, temperature, shading and fault conditions. The data were preprocessed and evaluated using k-fold cross-validation and hold-out testing. Bayesian optimization was used for classifier hyperparameter selection. The best-performing configurations were then selected by considering both classification accuracy and prediction speed. The authors also performed experimental verification using a physical PV setup.

TASK 2: METHODOLOGY ANALYSIS

2(a) ML Techniques, Dataset and Data Preparation

Step 1: PV System Used

The researchers considered a **4 kW grid-connected photovoltaic array** consisting of four parallel PV strings, with four series-connected PV modules in each string. The system contains power and control circuits and an MPPT controller.

The purpose of constructing this model was to reproduce different operating conditions and generate data representing both normal and faulty PV operation.

Step 2: Fault Conditions Considered

The study considers healthy and several faulty conditions. The major conditions include:

1. Healthy PV-array operation

2. Arc fault
3. MPPT unit failure
4. Line-to-line fault
5. Open-circuit fault
6. Partial/objective/subjective shading

The fault scenarios were generated under different environmental and electrical conditions.

Step 3: Dataset

A total of **14,100 samples** were collected from the MATLAB/Simulink PV-system model.

The dataset consisted of:

Condition	Number of Samples
Healthy condition	564
Arc fault	2,820
MPPT unit failure	2,820
Line-to-line fault	2,820
Open-circuit fault	2,820
Shading	2,256
Total	14,100

The dataset was generated using different values of solar irradiance, temperature, shading and fault impedance. The paper reports solar irradiance from approximately **50 to 1200 W/m²** and temperature from **0°C to 55°C**.

Step 4: Input Features

Four main input quantities were used by the fault detection and diagnosis system:

Feature	Description
Solar Irradiance	Amount of solar radiation received by the PV array
Temperature	Operating temperature
PV Array Voltage	Voltage measured from the PV array
PV Array Current	Current measured from the PV array

The voltage and current measurements were also used to calculate additional quantities such as PV output power and the gamma parameter.

Therefore, the input vector can be represented as:

$$\mathbf{X} = [\mathbf{G}, \mathbf{T}, \mathbf{VPV}, \mathbf{IPV}]$$

where:

- **G** = solar irradiance
- **T** = temperature
- **VPV** = PV-array voltage
- **IPV** = PV-array current

Step 5: Data Preprocessing

The authors performed preprocessing before applying machine-learning algorithms.

The major preprocessing operations were:

1. Converting raw measurements into a useful format.
2. Handling missing values.
3. Removing unsuitable records.
4. Converting categorical class information into numerical labels.
5. Applying normalization or standardization where required.

Standardization was particularly applied to **KNN and SVM**, because these algorithms are sensitive to feature scales. Decision Trees do not require the same type of scaling because tree decisions are based on threshold comparisons.

Step 6: Machine-Learning Models

The authors compared three main machine-learning techniques.

A. Decision Tree

Decision Tree is a supervised learning algorithm that recursively divides data according to feature conditions.

The study investigated different splitting criteria, including:

- Gini Index
- Twoing Rule
- Deviance

The Decision Tree models were evaluated separately for the detection and diagnosis stages.

B. K-Nearest Neighbors

KNN classifies a new sample according to the classes of its nearest training samples.

The researchers investigated different distance metrics and weighting methods. Since KNN depends on distances between samples, standardization of input features was applied.

C. Support Vector Machine

SVM is a statistical learning method that finds a suitable decision boundary between classes.

The study evaluated several SVM kernel functions:

- Linear

- Quadratic
- Cubic
- Gaussian/RBF

For diagnosis, different multi-class approaches such as one-versus-one and one-versus-all were investigated.

Step 7: Hyperparameter Optimization

Bayesian optimization was used to determine suitable hyperparameter values.

The basic process was:

Initial models → Hyperparameter search → Model evaluation → Best parameter selection → Final model

This helped the researchers obtain optimized versions of the classifiers rather than relying only on manually selected parameters.

Step 8: Model Validation

The researchers used two important evaluation procedures:

1. **k-fold cross-validation**
2. **Hold-out validation/testing**

The dataset was divided into training and testing portions. Approximately **65% of the data** were used for training and the remaining portion for testing. Cross-validation was used during model selection, while hold-out testing was used to evaluate the final model's generalization performance.

2(b) Mapping to CO4 Topics

CO4 Topic	Connection with the Article
Inductive Learning	The models learn general classification rules from labelled PV-system examples and use them to classify unseen samples.
Decision Trees	Decision Tree models using Gini Index, Twoing Rule and Deviance are directly investigated.
Statistical Learning	SVM is used as a statistical learning classifier and different kernel functions are compared.
Reinforcement Learning	Not directly used in the article. The article is mainly based on supervised learning.

Methodological Strength

A major strength is the **systematic comparison of multiple machine-learning classifiers and their configurations**. Instead of using only one model, the researchers compared Decision Trees, KNN and SVM and optimized their parameters. They also used both cross-validation and independent hold-out testing. This provides stronger evidence than reporting only training

accuracy.

Another strength is the use of both **simulation and experimental validation**, which increases the practical relevance of the research.

Methodological Limitation

The main limitation is that a large part of the dataset was **generated using a MATLAB/Simulink PV-system model** rather than being collected entirely from real-world solar farms. Simulated data can represent controlled fault scenarios well, but actual PV plants may contain additional noise, sensor errors, environmental variations and unexpected fault combinations.

Another limitation is the relatively small number of input features. Real PV installations may provide many additional measurements that could help distinguish complex faults.

TASK 3: RESULTS AND CRITICAL EVALUATION

3(a) Key Results

The researchers reported very high classification accuracy for the **fault detection stage**, while the more difficult **fault diagnosis stage** produced lower accuracy because distinguishing individual fault types is more challenging.

Decision Tree Results

Model	Application	CV Accuracy (%)	Test Accuracy (%)
M1 – Gini	Detection	99.37	99.73
M2 – Twoing	Detection	99.55	99.73
M3 – Deviance	Detection	99.68	99.79
M4 – Gini	Diagnosis	82.95	85.30
M5 – Twoing	Diagnosis	83.08	85.68
M6 – Deviance	Diagnosis	83.70	85.68

The best Decision Tree detection model achieved a **test accuracy of 99.79%**, while the best diagnosis models achieved **85.68%** test accuracy.

SVM Results

SVM Model	Kernel	Application	CV Accuracy (%)	Test Accuracy (%)
M15	Linear	Detection	99.97	99.97
M16	Quadratic	Detection	99.97	100.00
M17	Cubic	Detection	99.98	100.00
M18	Gaussian/RBF	Detection	99.98	100.00
M19	Linear	Diagnosis	76.24	75.38

M20	Quadratic	Diagnosis	82.64	81.73
M21	Cubic	Diagnosis	82.49	82.41
M22	Gaussian/RBF	Diagnosis	89.37	89.84

The optimized SVM models produced extremely high detection accuracy, reaching **100% test accuracy** for several configurations. For diagnosis, the best reported SVM model achieved **89.84% test accuracy** using a Gaussian/RBF kernel.

Overall Comparison

Technique	Best Detection Test Accuracy	Best Diagnosis Test Accuracy
Decision Tree	99.79%	85.68%
KNN	100.00%	89.84%
SVM	100.00%	89.84%

The results show an important difference between **fault detection** and **fault diagnosis**. Detecting whether the system is faulty is easier than determining exactly which fault has occurred. This explains why the detection accuracies are close to 100%, while diagnosis accuracies are considerably lower.

The article reports accuracy and misclassification cost as major performance measures rather than emphasizing F1-score or AUC-ROC. Therefore, F1-score and AUC-ROC should not be invented for this review.

3(b) Critical Evaluation of the Results

1. Are the Reported Metrics Realistic?

The reported detection accuracies of approximately 99–100% are possible because the dataset was generated under controlled simulation conditions. However, such extremely high values should be interpreted carefully.

The diagnosis results of approximately 85–90% are more realistic for a multi-class fault-identification problem because different PV faults can produce similar electrical characteristics.

The difference between detection and diagnosis demonstrates that the classification problem becomes more difficult when the number of fault classes increases.

2. Potential Dataset Bias

One potential bias is the use of simulated data. The researchers generated the main dataset using a MATLAB/Simulink PV model. Simulated signals are generally cleaner and more controlled than measurements obtained from real PV farms.

Another possible bias is that samples generated from similar operating conditions may be correlated. If highly similar samples are distributed between training and testing datasets, the test accuracy may appear higher than it would be on completely independent real-world data.

3. Class Distribution

The dataset is not perfectly balanced. For example, the healthy class contains only **564 samples**, whereas several fault categories contain **2,820 samples each**.

Therefore, accuracy alone may not completely describe performance across all classes. Additional measures such as:

- Precision
- Recall
- F1-score
- Confusion matrix
- Balanced accuracy
- ROC-AUC

would provide a more detailed evaluation.

4. Additional Experiments Required

The following experiments would strengthen the research:

1. Test the models on a completely independent real-world PV-farm dataset.
2. Introduce realistic sensor noise into the simulation.
3. Test different geographical and climatic conditions.
4. Evaluate performance under unseen combinations of faults.
5. Use precision, recall and F1-score for every fault class.
6. Perform repeated cross-validation with different random seeds.
7. Compare the proposed models with Random Forest, XGBoost and neural networks.
8. Measure real-time prediction latency on edge hardware.
9. Evaluate computational cost and memory requirements.
10. Perform statistical significance testing between the best classifiers.

These experiments would help determine whether the extremely high detection accuracy can be maintained outside the controlled experimental environment.

3(c) Real-World Applicability

The proposed machine-learning framework can be deployed in several real-world applications.

1. Solar Power Plants

The system can continuously monitor PV-array measurements and identify abnormal operating conditions.

2. Smart Grid Systems

PV fault diagnosis can be integrated with smart-grid monitoring systems to improve reliability

and reduce unexpected power losses.

3. Residential Solar Installations

Small rooftop PV systems can use an intelligent monitoring system to identify faults automatically and notify the owner or maintenance team.

4. Industrial Solar Farms

Large solar farms can use machine-learning-based fault detection to identify faulty strings or abnormal operating conditions before they result in significant energy losses.

5. Predictive Maintenance

The system can become part of a predictive-maintenance platform. Instead of waiting until a PV system fails, operators can receive early warnings and schedule maintenance.

Industry Scaling Challenges

Challenge	Explanation
Data availability	Large quantities of correctly labelled real-world fault data are difficult to collect.
Environmental variation	Solar irradiance and temperature vary continuously.
Sensor noise	Real sensors introduce measurement errors.
Fault rarity	Some dangerous faults occur very infrequently.
Computational requirements	Continuous monitoring of large solar farms requires efficient processing.
Model generalization	A model trained on one PV configuration may not perform equally well on another.
Cybersecurity	Connected monitoring systems must be protected from malicious manipulation.
Explainability	Maintenance engineers need understandable reasons for fault predictions.
Maintenance cost	Deploying sensors and computing devices throughout large solar farms can be expensive.

Therefore, although the article demonstrates strong potential, large-scale deployment would require further validation using real-world data.

TASK 4: REFLECTION AND LEARNING OUTCOME

4(a) Three Key Insights Gained from the Article

Insight 1: Decision Trees Can Learn Classification Rules from Data

The first important insight is that a Decision Tree can automatically learn classification rules by repeatedly splitting the data according to feature values.

This directly connects with the **Decision Tree concept in CO4**. The article demonstrates that different splitting criteria, such as Gini Index, Twoing Rule and Deviance, can produce different performance levels.

For example, the Decision Tree detection models achieved test accuracies ranging from **99.73% to 99.79%**, showing that the choice of splitting criterion can affect the final classifier.

CO4 Connection: Decision Tree learning and inductive classification.

Insight 2: Feature Preprocessing Is Important for Statistical Learning

The second insight is that preprocessing depends on the machine-learning algorithm.

The article standardizes the input features for KNN and SVM because these methods can be affected by differences in feature scales. Decision Trees do not require the same scaling because they use threshold-based comparisons.

This demonstrates that machine-learning performance is influenced not only by the algorithm but also by the way the input data are prepared.

CO4 Connection: Statistical learning, feature preprocessing and supervised learning.

Insight 3: Detection and Diagnosis Are Different ML Problems

The third important insight is that detecting a fault and identifying the exact fault are different levels of classification difficulty.

The models achieved nearly 100% accuracy for fault detection, while diagnosis accuracy was approximately 85–90%. This shows that identifying "normal versus faulty" is easier than distinguishing among several similar fault types.

This helped me understand that the number of classes, similarity between classes and quality of features can significantly influence classification performance.

CO4 Connection: Inductive learning, multi-class classification and model evaluation.

4(b) Proposed Novel Extension

Proposed Experiment: Real-Time Hybrid PV Fault Diagnosis Using Explainable Machine Learning

If I were to extend this research, I would develop a **real-time hybrid PV fault diagnosis system using real-world sensor data and explainable machine learning**.

The proposed experiment would collect voltage, current, solar irradiance and temperature measurements from several PV installations under different weather conditions. The dataset would contain both healthy and faulty conditions. Sensor noise and missing values would also be intentionally included to represent real-world conditions.

The original Decision Tree and SVM models could be compared with a modern ensemble method such as Random Forest or XGBoost. The models would be evaluated using accuracy, precision, recall, F1-score, confusion matrix and inference time.

An explainability technique could then be used to determine which features contributed most to each prediction. For example, the system could report that a combination of unusual PV voltage and current measurements contributed to the prediction of a particular fault.

Feasibility

The experiment is feasible because voltage, current, temperature and irradiance sensors are already commonly used in PV monitoring systems. Data can be collected using low-cost microcontrollers and data-acquisition devices. Machine-learning models can also be deployed on edge computing devices.

Expected Impact

The expected impact is improved reliability and practical applicability. Testing on real-world data would determine whether the high simulation-based accuracy can be maintained in actual solar installations. Explainable predictions would also increase the confidence of maintenance engineers in using the system.

The extended system could ultimately support **automatic fault detection, predictive maintenance, reduced downtime, improved energy generation and lower maintenance costs**.

CONCLUSION

The reviewed article presents a systematic machine-learning approach for detecting and diagnosing faults in photovoltaic arrays. The researchers compare Decision Tree, KNN and SVM classifiers and optimize their configurations using Bayesian optimization. A dataset containing 14,100 samples was generated under different irradiance, temperature, shading and fault conditions. The results show extremely high performance for fault detection, with several models achieving approximately 100% test accuracy, while fault diagnosis is more challenging and produces lower accuracy.

The study provides a strong practical example of **inductive learning, Decision Trees and statistical learning methods**, which are important components of CO4. Its major strength is the comparison of several algorithms using cross-validation and hold-out testing, together with experimental verification. However, the use of primarily simulated data means that further testing on large, diverse real-world PV datasets is required.

Overall, the article demonstrates how machine-learning techniques can convert raw PV measurements into useful fault classifications. It also shows that model selection, feature preprocessing, validation methodology and real-world testing are essential for developing reliable machine-learning systems.

REFERENCES

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